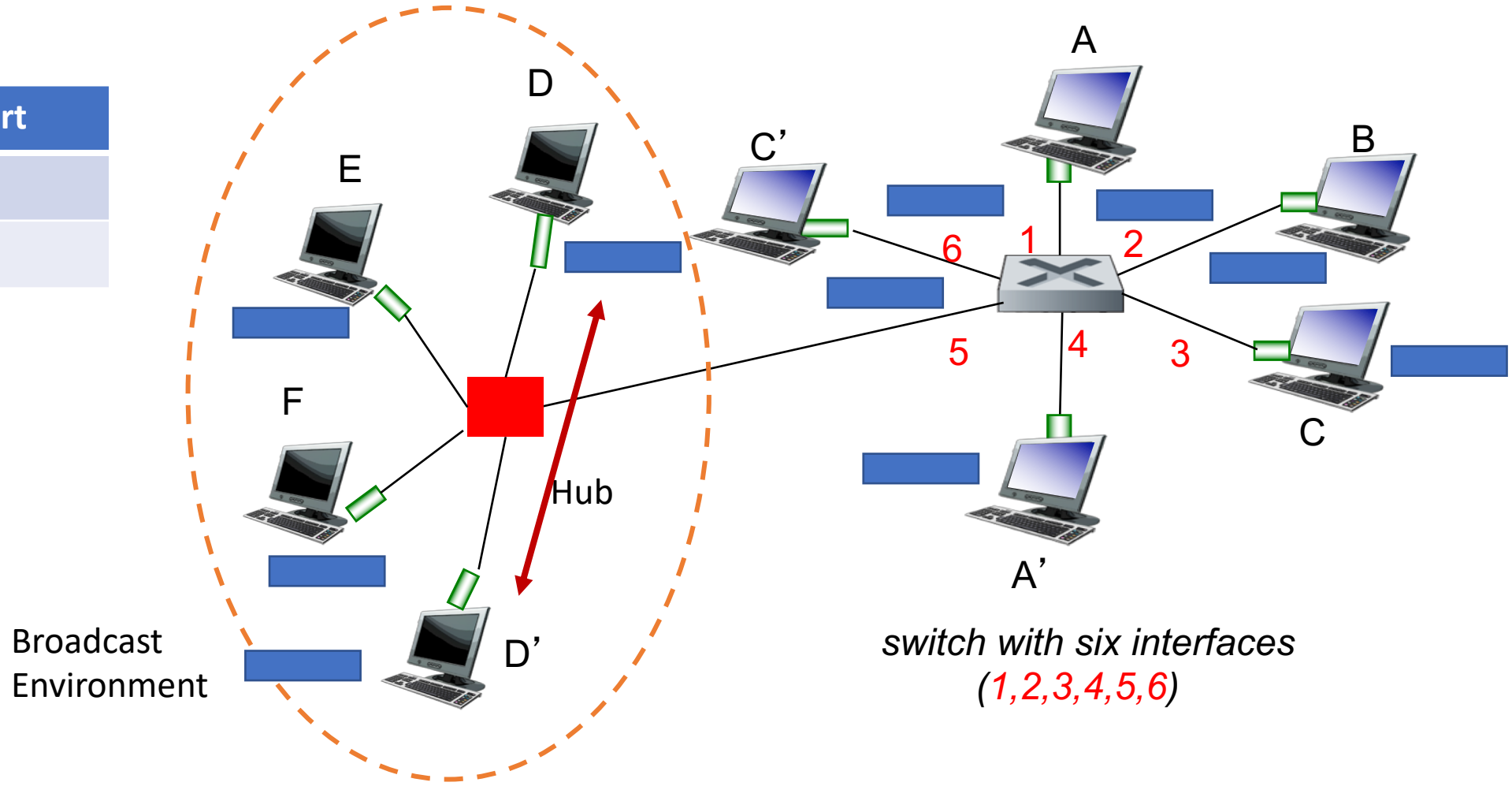


Switch & Broadcast Environment

Address	Port
D	5
D'	5



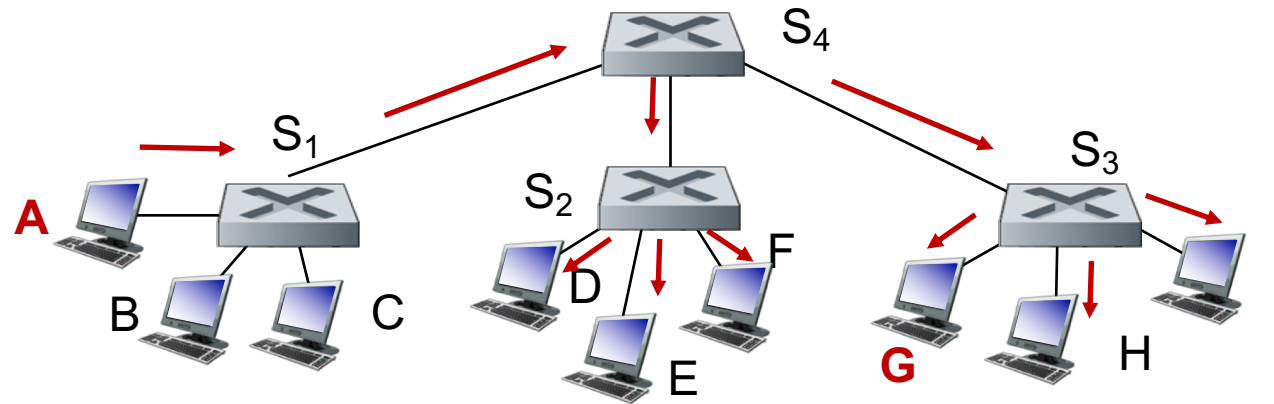
Switch: frame filtering/forwarding

when frame received at switch:

1. record incoming link, MAC address of sending host
2. index switch table using MAC destination address
3. if entry found for destination
 then {
 if destination on segment from which frame arrived (i.e., same segment)
 then drop frame
 (i.e, destination and source in same direction, no need to forward, see D-D' next slide)
 else forward frame on interface indicated by entry
 }
 else flood /* forward on all interfaces except arriving
 interface */

Interconnecting switches

self-learning switches can be connected together:

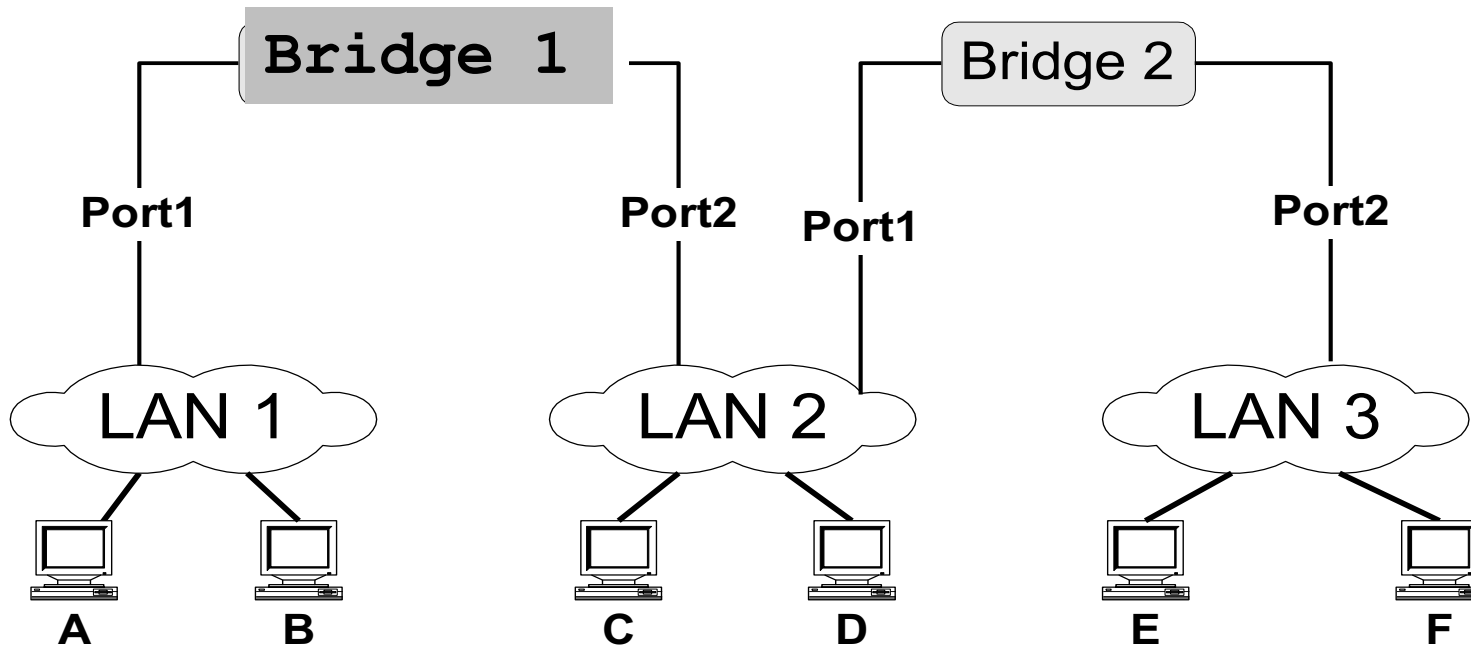


Q: sending from A to G - how does S₁ know to forward frame destined to G via S₄ and S₃?

A: self learning! (works exactly the same as in single-switch case!)

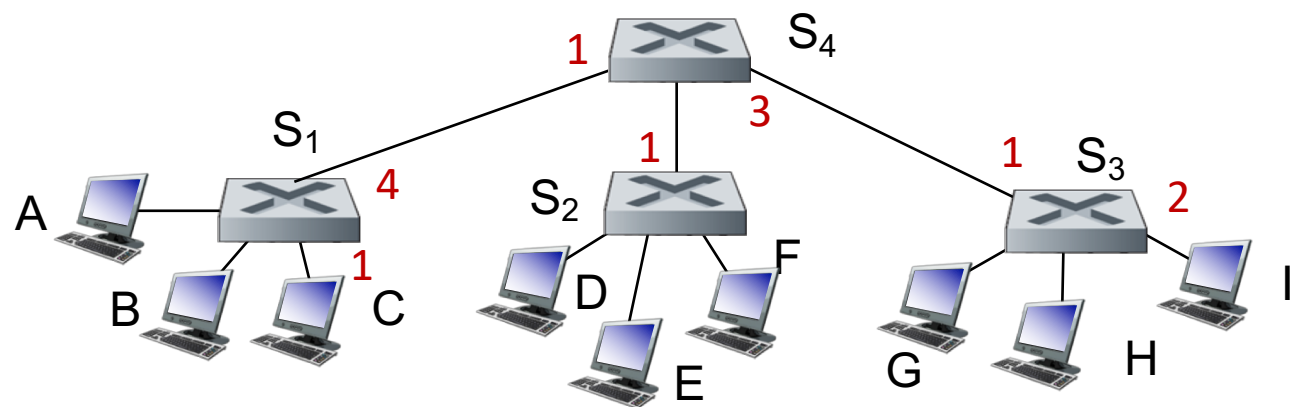
Example

- the following packets have been transmitted in the two bridge network below:
(Src=A, Dest=F), (Src=C, Dest=A), (Src=E, Dest=C)
- After the 3 transmissions: what have the bridges learned?



Self-learning multi-switch example

Suppose C sends frame to I, I responds to C



S1	
Address	Port
C	1
I	4

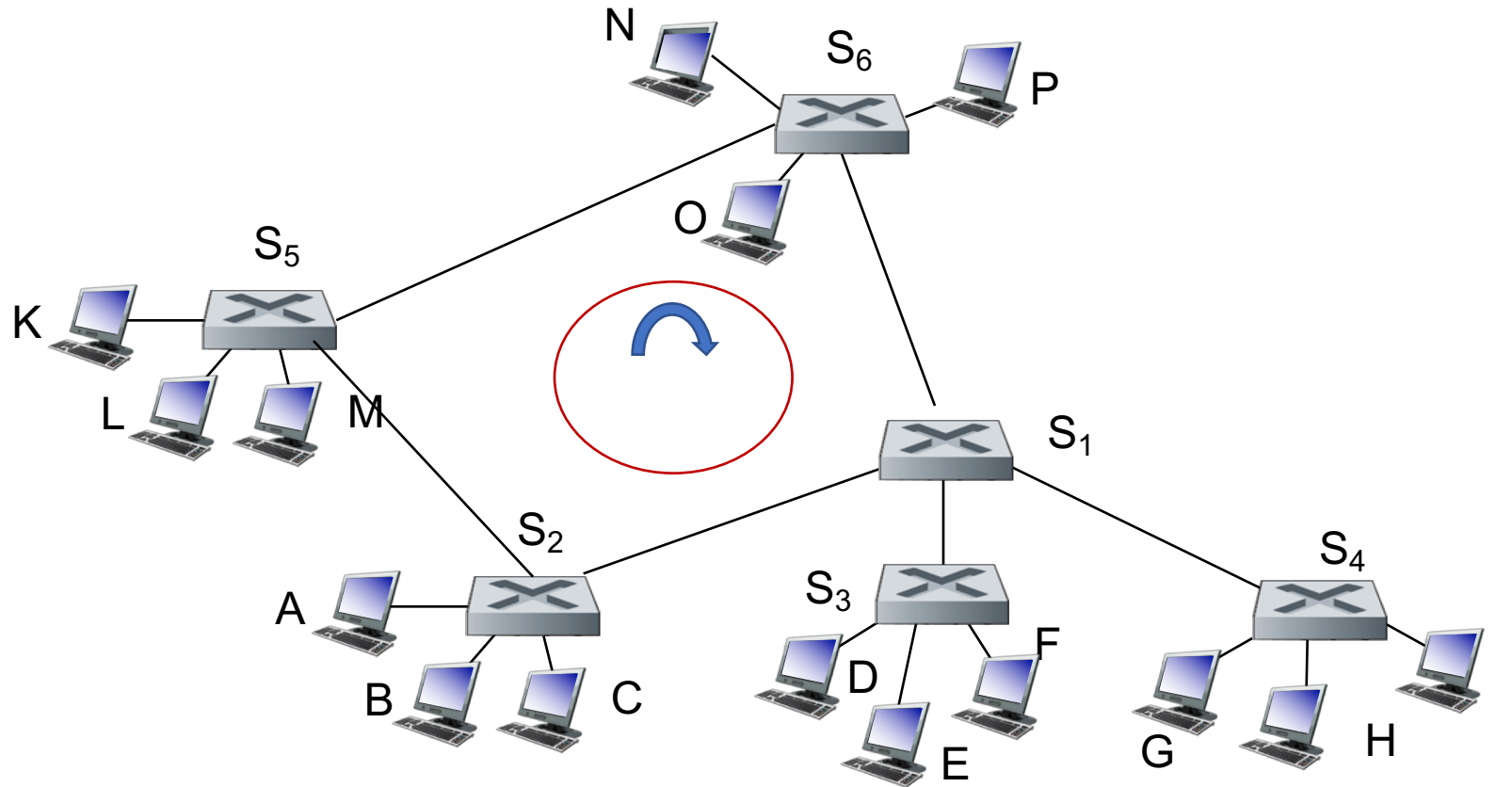
S4	
Address	Port
C	1
I	3

S3	
Address	Port
C	1
I	2

- Q: show switch tables and packet forwarding in S₁, S₂, S₃, S₄

S2	
Address	Port
C	1

Loops and Routing



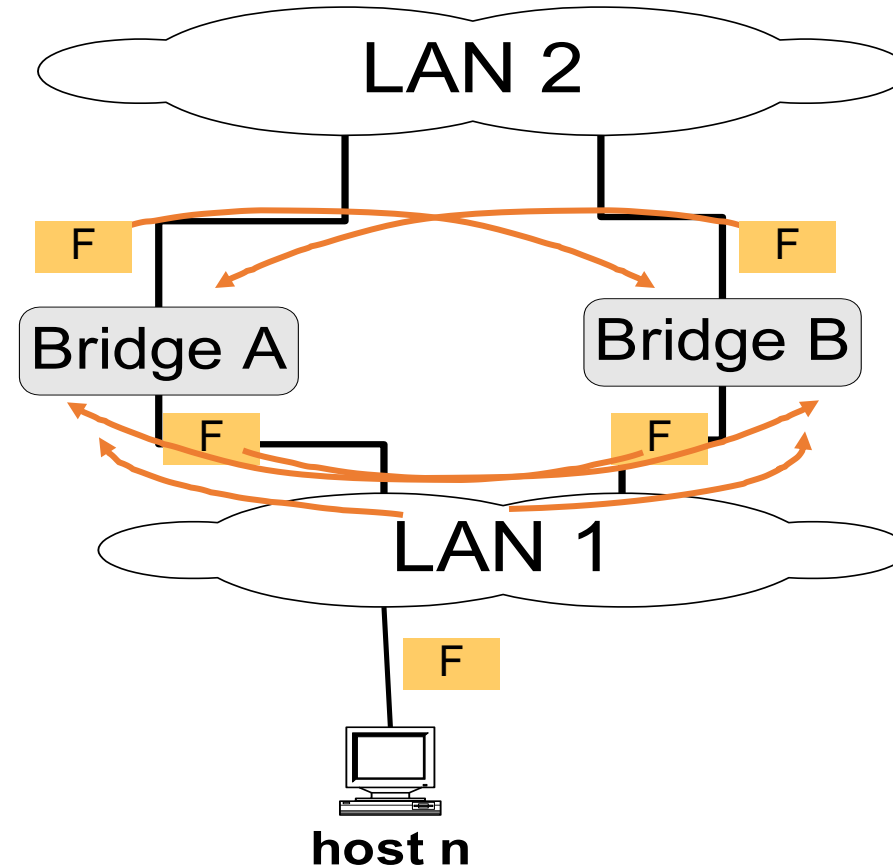
Danger of Loops

- Consider the two LANs that are connected by two bridges.
- Assume *host n* is transmitting a **frame F** with unknown destination.

What happens?

- Bridges A and B flood the frame to LAN 2 and enter LAN1 as source for host n.
- Bridge B sees F on LAN 2, and copies the frame back to LAN 1 as it has no destination for F.
- Bridge A does the same.
- The copying continues

Where's the problem? What's the solution? **Prevent looping**



Spanning Tree

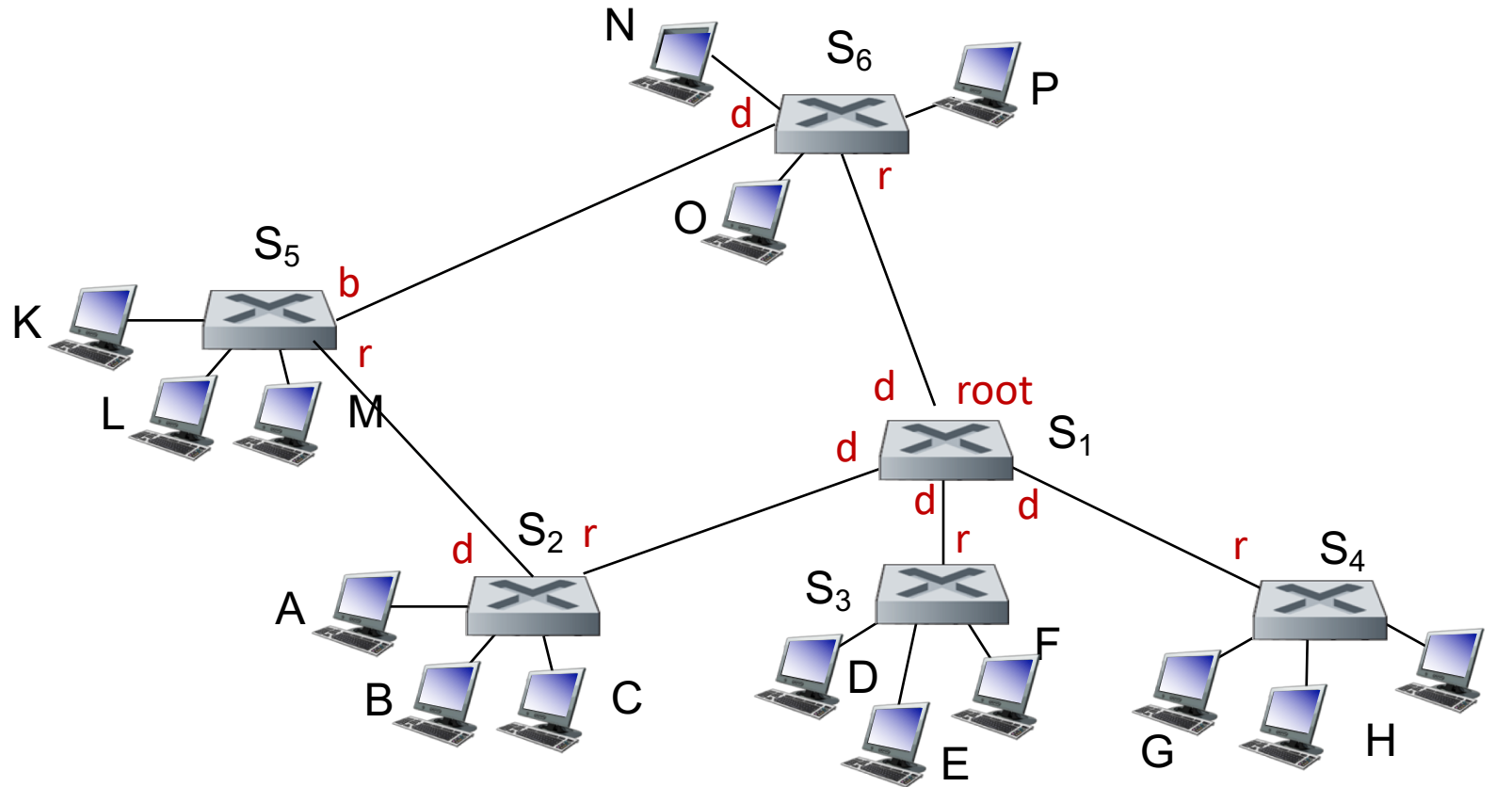
- a solution is to prevent loops in the topology
- IEEE 802.1d standardized an algorithm that builds and maintains a **spanning tree** in a dynamic environment
- switches exchange messages to configure the spanning tree
(**Configuration Bridge Protocol Data Unit** - Configuration BPDU)
- uses flooding, takes sometime to converge
- one bridge elected the root, switches choose shortest path to root
- bridges that run 802.1d are called transparent bridges

BPDUs and Spanning Tree

with the help of BPDUs, switches:

- elect a single switch as the **root switch** (switch with lowest ID)
- calculate the distance of the **shortest path** to the root switch
- a shared medium (broadcast LAN) with multiple switches, picks a **designated switch**, which is the switch with shortest path to root
- ties in shortest paths are broken by choosing a path leading to a switch with lower ID
- each switch has a **root port**, it is the port that leads to the root via the shortest path
- a designated switch will have a **designated port** that will be used for frame forwarding (part of spanning tree)
- switches will forward packets on root and designated ports only
- some ports on a switch will be **blocked**, i.e., not in spanning tree, will not forward data (prevent looping)
- root and designated ports are **active** – **forward** traffic

Spanning Tree example

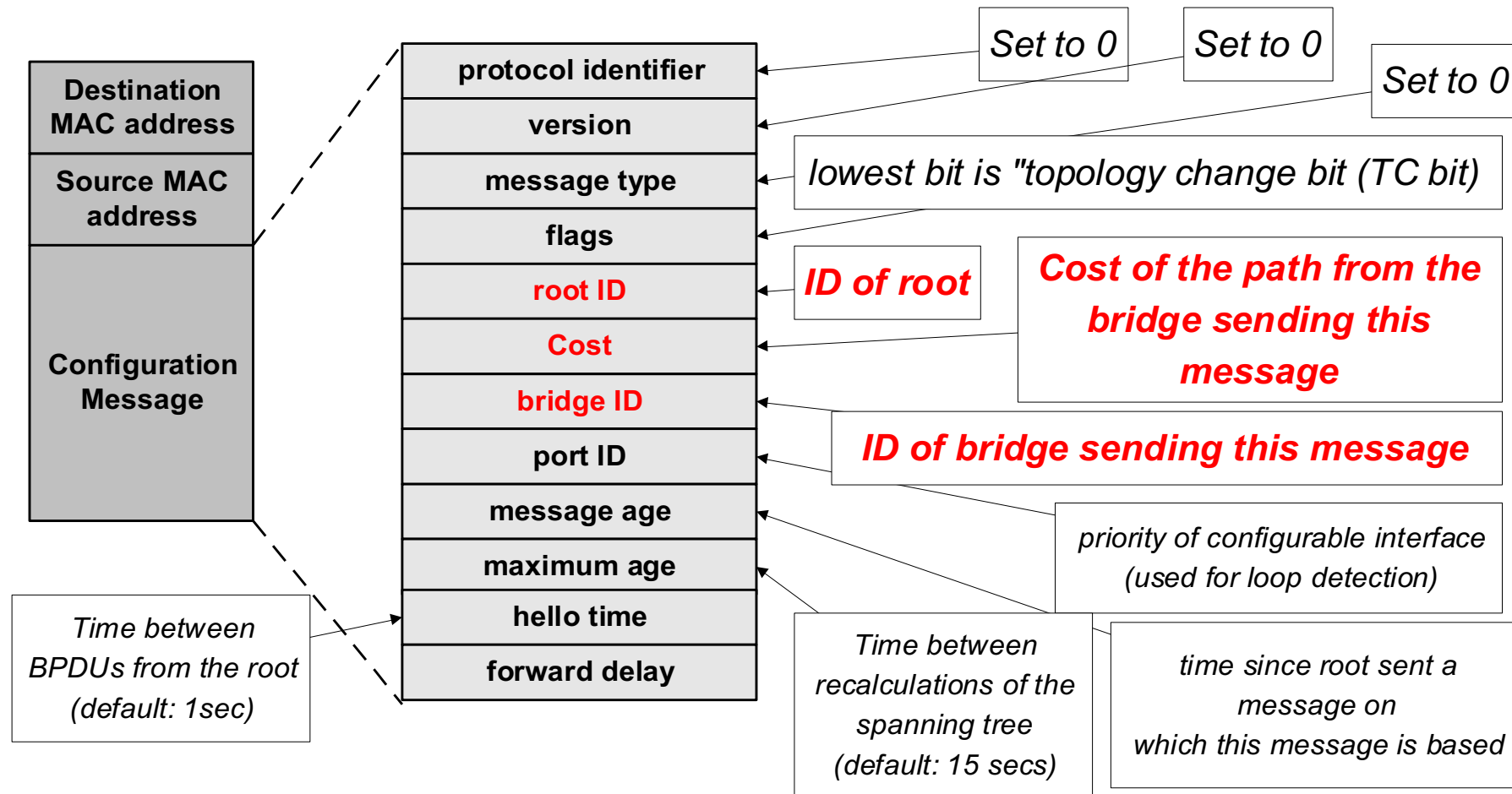


What do the BPDUs do?

with the help of the BPDUs, bridges can:

- elect a single bridge as the **root bridge**.
- calculate the distance of the **shortest path** to the root bridge
- each LAN can determine a **designated bridge**, which is the bridge closest to the root on a shared medium (e.g., Ethernet). The designated bridge will forward packets towards the root bridge.
- each bridge can determine a **root port**, the port that gives the best path to the root.
- select ports to be included in the spanning tree (**Active and Passive/Blocked**).

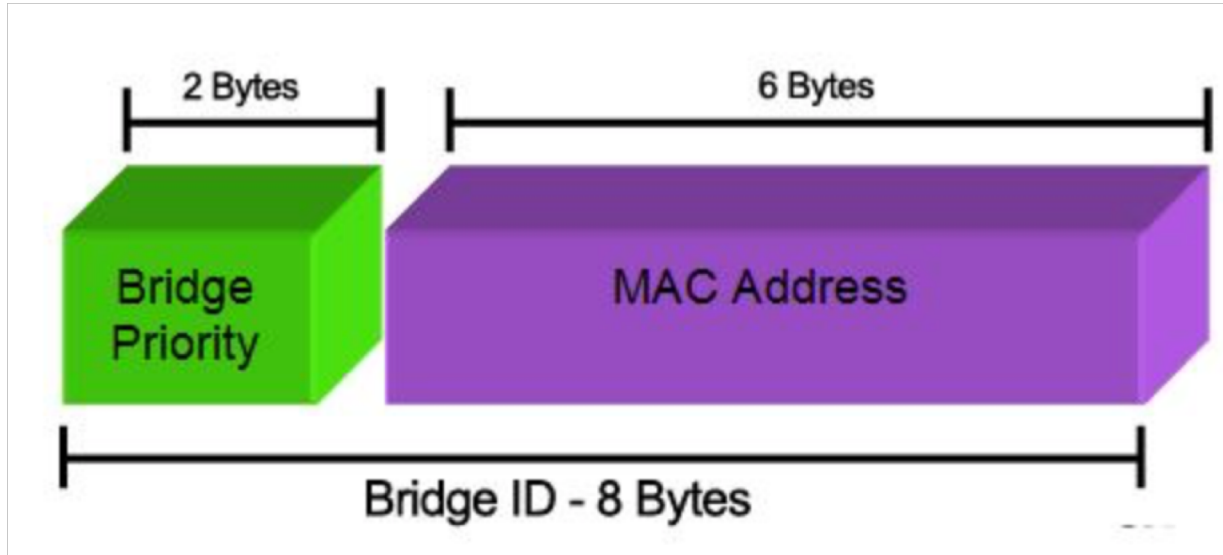
Configuration BPDUs



Concept

- Each bridge as a unique identifier:

Bridge ID = <priority level.MAC address>



- Note that a bridge has several MAC addresses (one for each port), but only **one ID**. It picks the **lowest MAC address**
- Each port within a bridge has a unique identifier - **port ID**.
- Root Bridge:** The bridge with the lowest ID is the root of the spanning tree. I.e., lowest MAC and priority level.
- Root Port:** Each bridge has a root port which identifies the next hop from that bridge to the root. This port is identified during the spanning tree process

Building a Spanning Tree

- **Root Path Cost:**

For each bridge, the cost of the min-cost path to the root.

Usually it is measured in #hops to the root

- **Designated Bridge, Designated Port:**

- Single bridge on a LAN that provides the minimal cost path to the root bridge for this LAN:

- if two bridges on a LAN have the same cost, select the one with lowest ID
- if the min-cost bridge has two or more ports on the LAN, select the port with the **lowest port ID**